Learning Journey Assistant — Requirements derived from owner meeting

Source: recorded meeting with Dr Scott Mann (project owner), 11 August 2026. Attendees: Allan Campton, Istiaque Bhuiyan, Scott Mann. Status: draft for team review, then Jira import.

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1. Scope statement

The system ingests subject intended learning outcomes (SILOs) and student assessment results across a multi-year subject suite, semantically links SILOs into labelled competency categories, aggregates student performance through those categories, identifies competency gaps, and recommends targeted interventions.

Primary trigger cadence is **between semesters** (end-of-semester results feed the next semester's study plan). In-semester triggering is explicitly a later phase.

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2. Data supplied by the owner

Single Excel workbook, distributed via MS Teams and the LMS, covering three subjects:

Code	Level	Role
OOF	1	Object-oriented fundamentals, core
ALG	2	Algorithmics, core
CAP	3	Capstone, core to all undergraduates

Tab: Assessment map

Field	Priority	Notes
`subject_code`	key	
`assessment`	key	
`silo`	key	Assessment to SILO is one-to-many
`silo_summary_text`	key	Free text; the semantic input
`weighting`	low	Retain, do not analyse
`contribution` (group/individual)	low	Secondary consideration
`early_assessment` flag	low	Owner: immaterial
`hurdle` flag	low	Owner: not required

Tab: Student summary

`student_id`, per-subject outcome, final grade, average, performance band, at-risk flag. The at-risk rule is threshold-based but the exact threshold is unconfirmed — **open question for the owner**.

Tab: Results

`student_id`, `subject_code`, `assessment`, `score`, `feedback_comment`, `silo`. Weighted score present but low priority. `student_id` joins across all three tabs.

Known data constraint

Feedback comments in the supplied extract come from Moodle's automated rubric feedback and are near-identical canned sentences per band. Bespoke assessor feedback is far more informative but carries institutional re-identification risk. The owner will attempt to supply a sanitised extract. **This limitation must be recorded in the final handover report as a stated constraint on outcomes.**

Available on request

Standard course outlines / course maps for the Computer Science, IT and Cyber degrees.

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3. Functional requirements

FR-1 — Data ingestion

ID	Requirement	Priority
FR-1.1	Parse the supplied workbook's three tabs into a normalised relational schema.	Must
FR-1.2	Model assessment-to-SILO as one-to-many; never collapse to one-to-one.	Must
FR-1.3	Ingestion is idempotent and re-runnable; re-import of the same source produces no duplicate rows.	Must
FR-1.4	Ingestion sits behind a source adapter interface so the spreadsheet and the live Moodle database populate the same canonical model.	Must
FR-1.5	Pseudonymise `student_id` at the ingest boundary; the raw identifier never leaves the ingest layer.	Must
FR-1.6	Reject and report malformed rows rather than silently dropping them; emit an ingest quality report.	Should
FR-1.7	Ingest a course map (subject sequence per degree) when supplied by the owner.	Should

FR-2 — SILO registry

ID	Requirement	Priority
FR-2.1	Each SILO has a stable identity of `(subject_code, silo_index)` plus verbatim summary text.	Must
FR-2.2	Store subject level (½/3) and provenance (handbook, SLG, supplied workbook).	Must
FR-2.3	SILO text is immutable; a change of wording creates a new version rather than overwriting.	Must
FR-2.4	Support a suite of 30+ subjects without schema change.	Must

FR-3 — Semantic linking (core capability)

ID	Requirement	Priority
FR-3.1	Compute a semantic similarity score for every cross-subject SILO pair, using embeddings and/or LLM reasoning over full sentence meaning.	Must
FR-3.2	Keyword or lexical overlap must not be the production mechanism. It may be implemented only as a baseline for evaluating the semantic approach.	Must
FR-3.3	Apply a configurable similarity threshold; pairs below it produce no edge. The threshold value in force is recorded with every generated map.	Must
FR-3.4	Every retained edge stores: score, natural-language rationale, model identifier, prompt version, timestamp.	Must
FR-3.5	An academic can accept, reject or adjust any edge through the interface. Human overrides persist and take precedence over machine output on regeneration.	Must
FR-3.6	Expose the SILO graph as an exportable network structure (nodes = SILOs, edges = semantic links) suitable for network visualisation.	Must
FR-3.7	Recomputation is incremental: adding one subject does not require reprocessing all existing pairs.	Should

FR-4 — Competency categorisation

ID	Requirement	Priority
FR-4.1	Cluster semantically linked SILOs into competency groups.	Must
FR-4.2	Generate a concise human-readable label and description for each competency (e.g. "object-oriented modelling", "written communication").	Must
FR-4.3	Allow a SILO to belong to more than one competency, with membership strength.	Must
FR-4.4	Each competency resolves to: member SILOs, containing subjects, subject levels, mapped assessments.	Must
FR-4.5	Competency labels are editable and lockable by an academic.	Should

FR-5 — Performance aggregation and gap detection

ID	Requirement	Priority
FR-5.1	Compute a per-student competency score by aggregating assessment scores through assessment to SILO to competency.	Must
FR-5.2	The aggregation function is pluggable; default is the unweighted mean, since the owner deprioritised assessment weighting.	Must
FR-5.3	Detect *intra-student* deficiency: competency scores materially below that student's own overall mean. This is the owner's stated primary signal — variability within a discipline, not raw failure.	Must
FR-5.4	Detect *inter-student* deficiency: competency scores materially below the cohort distribution.	Must
FR-5.5	Track competency scores longitudinally across levels 1, 2 and 3 to expose trajectory.	Must
FR-5.6	Every flagged gap drills down to the specific assessments, SILOs and scores that produced it. No unexplained flags.	Must
FR-5.7	Deficiency thresholds are configurable, not hardcoded.	Must
FR-5.8	Reproduce and expose the existing at-risk band from the student summary tab alongside computed gaps, for comparison.	Should

FR-6 — Feedback comment analysis (team differentiator)

ID	Requirement	Priority
FR-6.1	Classify each feedback comment as templated or bespoke, e.g. by near-duplicate detection across the cohort.	Must
FR-6.2	For bespoke comments, extract concept-level deficiencies (for example "polymorphism", "recursion", "requirements	Must

ID	Requirement	Priority
	traceability”) and attach them to the relevant SILO and competency.	
FR-6.3	Degrade gracefully: when only templated feedback exists, gap detection falls back to score-based signals alone, and the reduced confidence is surfaced.	Must
FR-6.4	Provide a synthetic feedback generator for development and testing, with controlled variability. Synthetic records are flagged at row level and can never be mixed into production analytics.	Should
FR-6.5	Report the templated/bespoke ratio per subject as a data-quality metric.	Should

FR-7 — Intervention

ID	Requirement	Priority
FR-7.1	For each identified gap, produce ranked intervention recommendations.	Must
FR-7.2	Support at minimum three intervention classes: elective or subject suggestions that backfill the competency, curated external learning resources, and generated practice quizzes or study material.	Must
FR-7.3	Recommendations are contextualised against the student’s forward study plan and the course map when available.	Should
FR-7.4	Every recommendation cites the gap and the evidence that triggered it.	Must
FR-7.5	The trigger mechanism is decoupled from the end-of-semester batch, so in-semester triggering (e.g. after a week-4 early assessment) can be enabled later without redesign.	Must

FR-8 — SILO quality critique

ID	Requirement	Priority
FR-8.1	Assess each SILO’s text for clarity, specificity and measurability, including presence of an observable verb and whether it fails to link semantically to anything in the suite.	Should
FR-8.2	Produce a report for subject coordinators listing low-quality SILOs, the justification, and a suggested rewording.	Should

Explicitly requested by the owner as a valued secondary outcome.

FR-9 — Export for research

ID	Requirement	Priority
FR-9.1	Export the competency map, per-student gap analysis and intervention log in machine-readable form (CSV and JSON).	Must
FR-9.2	Exports are de-identified by default; re-identification requires a separate, audited step.	Must
FR-9.3	Exports carry a snapshot timestamp and the configuration (model, thresholds, prompt versions) used to produce them, so before/after and A/B comparisons are valid.	Must

This closes the owner must-have previously flagged as absent from the roadmap.

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4. Non-functional requirements

ID	Requirement
NFR-1	Scale from 3 subjects to 30+. Pairwise comparison is quadratic; the design must use vector retrieval to shortlist candidate pairs before any LLM adjudication, and must cache embeddings. A naive all-pairs LLM pass is not acceptable.
NFR-2	Privacy. Pseudonymised identifiers only; no student-identifying text in prompts, logs or traces.
NFR-3	Reproducibility. Model version, prompt version and thresholds are persisted with every generated artefact. Adjudication runs at temperature 0.
NFR-4	Provider abstraction. The LLM provider is swappable behind an interface, with a deterministic offline stub for CI and for demonstration without network access.
NFR-5	Cost and rate limiting. Token spend per full recomputation is bounded and reported.
NFR-6	Explainability. No automated claim is presented to an academic or student without traceable evidence.
NFR-7	Local-first. The full pipeline runs against the local Dockerised Moodle instance with no institutional network dependency.
NFR-8	The data-cleansing constraint on feedback comments is documented in the final report as a limitation on outcome validity.

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5. Open questions for the owner

1. What is the exact threshold or rule behind the at-risk performance band?
2. Can a sanitised extract of bespoke assessor feedback be supplied, and on what timeline?
3. Please supply the course maps for the Computer Science, IT and Cyber degrees.
4. Is there a preferred or prohibited LLM provider from a La Trobe data-governance standpoint?
5. Is there an existing institutional competency framework the labels should align to, or are the team free to derive labels from the data?
6. Confirm a recurring weekly meeting slot.

6. Governance notes

- The same dataset has been supplied to all teams. Baseline understanding may be shared; technical approach and implementation are to be kept within the team from this week onward.
- The owner has offered weekly availability. The team is to propose a time.
- The project is intended for production use over multiple years, with handover to subsequent student cohorts. Handover artefact naming must match the owners' terminology.